

Kunalan Kevin Subagaran

kunalan.k.subagaran@gmail.com | [linkedin.com/in/knlInks/](https://www.linkedin.com/in/knlInks/) | github.com/KnlInKS

Languages & Technologies

Languages: TypeScript, JavaScript, Go, Python, Rust, SQL, Bash, Java, C

Backend & Platform: Temporal, Node.js, REST, GraphQL, gRPC, WebSockets

Infrastructure: Kubernetes, Docker, Helm, Terraform, Pulumi, AWS, GCP

Data & Security: PostgreSQL, Redis, Kafka, ClickHouse, Quickwit, Prometheus, gVisor, nsjail

Experience

Retool

July 2023 - Present

Software Engineer

San Francisco, CA

- Owned Temporal-backed orchestration for Retool Agents, supporting **50K+ durable workflows/month** across retries, cancellation, timeouts, and idempotent state transitions; reduced **[TODO: stuck workflows/manual recoveries from X to Y]**
- Shipped hardened sandboxing for untrusted agent tool execution using **nsjail** and **gVisor**, isolating filesystem, network, process, and syscall access; unblocked **[TODO: customer/workload/product launch]** while reducing **[TODO: execution failures/security review time from X to Y]**
- Improved response times on core product routes from **2s to 200ms** by introducing caching, eliminating inefficient async patterns, and optimizing PostgreSQL queries
- Led migration from Datadog to a self-hosted observability stack, saving **\$750K+** annually while preserving metrics, logs, traces, and alerting for **[TODO: service count or ingestion volume]**
- Built telemetry for on-prem deployments, surfacing customer health, usage, and deployment diagnostics to reduce enterprise support investigations by **[TODO: count/week from X to Y or verified %]**
- Built endpoint ownership mapping across **100+ services and routes**, routing incidents to accountable owners and reducing on-call triage from **[TODO: X minutes to Y minutes]**
- Drove platform rollout and deployment customizations for Apple and NBCU, unblocking **\$2.5M+ ARR** through enterprise observability, deployment, and integration support

Snapchat

May 2022 - August 2022

Software Engineer Intern

Toronto, ON

- Implemented hybrid encryption workflows on AWS and GCP for sensitive user data, integrating cloud KMS and envelope encryption across production data-processing paths
- Built a backend service handling **1.7M+ requests/day** for model-training data processing, improving input quality for AR filter training pipelines

Bonsai

Jan 2021 - August 2021

Software Developer Intern

Toronto, ON

- Engineered an external order-fulfillment system that processed over **\$1.6M** in volume within 2 months by integrating partner APIs, inventory sync, and checkout flows
- Built partner-facing dashboard workflows in React and TypeScript for search, filtering, exports, and product verification, reducing **[TODO: merchant ops time from X to Y]**
- Led migration of an embedded e-commerce widget from React to Preact, reducing client-side bundle size from **[TODO: X KB to Y KB]** and improving page load performance

Education

Queen's University

September 2019 - May 2023

Bachelor of Computing (Honours) in Software Design

Kingston, ON

- Awards:** Ratehub.ca Award in Computing, Dean's Honour List × 4